1.Converting an Integer into Decimals

In [8]:
```
import decimal
integer = 10
print(decimal.Decimal(integer))
print(type(decimal.Decimal(integer)))
```

```
10
<class 'decimal.Decimal'>
```

2. Converting a String of Integers into Decimals

In [11]:
```
import decimal
string = '12345'
print(decimal.Decimal(string))
print(type(decimal.Decimal(string)))
```

```
12345
<class 'decimal.Decimal'>
```

3.Reversing a String using an Extended Slicing Technique

In [14]:
```
string = "Chinmayee Nayak"
print(string[::-1])
```

```
kayaN eeyamnihC
```

4.Reverse a string without using slicing

Write a Python program to reverse the string "Chinmayee Nayak" without using slicing.**

Solution 1: Using a for loop

In [18]:
```
string = " Chinmayee Nayak"
reversed_string = ""

for char in string:
    reversed_string = char + reversed_string
print(reversed_string)
```

```
kayaN eeyamnihC
```

Solution 2: Using reversed() function

In [24]:
```
string = "Chinmayee Nayak"
reversed_string = "".join(reversed(string))

print(reversed_string)
```

```
kayaN eeyamnihC
```

Solution 3: Using a While Loop

In [27]:
```
string = "Chinmayee Nayak"
i = len(string) - 1
reversed_string = ""

while i >= 0:
    reversed_string += string[i]
    i -= 1

print(reversed_string)
```

```
kayaN eeyamnihC
```

5.Counting Vowels in a Given Word

In [1]:
```
vowel =['a','e','i','o','u']
word="Programming"
count=0
for character in word:
    if character in vowel:
        count += 1
print(count)
```

```
3
```

Python program to count number of vowels in a string

Algorithm

Step 1:- Start. Step 2:- Take user input. Step 3:- Initialize count variable. Step 4:- Iterate through the string to find number of vowels. Step 5:- Check if the alphabet of the string lies under the group of vowels. Step 6:- If TRUE increment count by 1. Step 7:- Print count. Step 8:- End.

In [4]:
```
#take user input
String = input('Enter the string:')
count = 0
#to check for less conditions
#keep string in lowercase
String = String.lower()
for i in String:
    if i == 'a' or i == 'e' or i == 'i' or i == 'o' or i == 'u':
        # If True
        count+=1
# Check if any vowel found
if count == 0:
    print('No Vowels Found')
else:
    print('Total Vowels are:' + str(count))
```

```
Total Vowels are:6
```

6. Python Program to Add Two Numbers

Soultion-1: (With Pre defined variables)

In [3]:
```
num1 = 30
num2 = 40
print("the sum of given two number is",num1 + num2)
```

```
the sum of given two number is 70
```

or

In [6]:
```
num1 = 30
num2 = 40
sum = num1 + num2
print("the sum of given two number is",sum)
```

```
the sum of given two number is 70
```

Solution-2 : (With user-inputs)

In [9]:
```
num1 = float(input("enter a number here :"))
num2 = float(input("enter another number here :"))

sum = num1 + num2
print("the sum of the provided two numbers is ", sum)
```

```
the sum of the provided two numbers is  100.0
```

7.HELLO WORLD PROGRAM IN PYTHON - First Python Program

In [13]:
```
# python program to print "HELLO WORLD"
print("hello world")
```

```
hello world
```

8.Find & Calculate Square Root Program in Python

Soultion-1 : (Using Exponentiation)

In [25]:
```
#num = 64
num1 = int(input("Enter a number here: "))
sr = num1**(1/2)
print("The square root of given number is ",sr)
```

```
The square root of given number is  7.0
```

Soultion-2 : (Using Math Module)

In [29]:
```
import math
num = int(input("Enter a number here: "))
sr = math.sqrt(num)
print("The square root of the given number is",sr)
```

```
The square root of the given number is 8.0
```

9.How to Calculate Area of Triangle - Python Program

In [34]:
```
height = float(input("Enter the height of the triangle:"))
base = float(input("Enter the base of the triangle:"))

area = (0.5)*base*height

print("The area of triangle is ",area)
```

```
The area of triangle is  34.375
```

10.Python Program - Swap Two Variables in Python

Solution-1:(Using thirD Variable)

In [38]:
```
x = 13
y = 12

temp = x
print("The value of temp variable is",temp)

x = y
print("The value of x is ",x)

y = temp
print("The value of y is",y)
```

```
The value of temp variable is 13
The value of x is  12
The value of y is 13
```

Soultion-2: (Without using third variable)

In [41]:
```
x = 12
y = 13

x,y = y,x
print("The value of x is", x)
print("The value of y is", y)
```

```
The value of x is 13
The value of y is 12
```

11.Python Program to Convert Kilometers to Miles

km = float(input("Enter your value in kms: "))

miles = (0.621371)*km print(km,"kms will be",miles,"miles")

12.Python Program to check if a Number is Positive ,Negative or Zero

Soultion:(Using Conditional Statements)

In [54]:
```
num = int(input("Enter a Number Here:"))

if num > 0:
    print("It is a Positive Number")
elif num == 0:
    print("it is zero")
else:
    print("It is Negative Number")
```

```
it is zero
```

13. Python Program to Check if a Number is Odd or Even?

In [61]:
```
num = int(input("Enter a number here:"))
if num % 2 == 0:
    print("It is an even number")
else:
    print("It is an odd number")
```

```
It is an even number
```

14.Python Program to Check Leap Year

In [7]:
```
year = int(input("Enter a year:"))
if (year % 400 == 0) and (year % 100 == 0):
    print(year,"Is a Leap Year")

elif (year % 4 == 0) and (year % 100 != 0):
    print(year,"Is a Leap Year")

else:
    print(year,"Is not a Leap Year")
```

```
2026 Is not a Leap Year
```

15.Python program to Find the Largest Among Three Numbers

Solution-1:(Using Conditional Statements)

In [11]:
```
num1 = 12
num2 = 56
num3 = 36

if ( num1 > num2 ) and (num1 > num3):
    print(num1,"is the largest number")
elif ( num2 > num1 ) and (num2 > num3):
    print(num2,"is the largest number")
else:
    print(num3,"is the largest number")
```

```
56 is the largest number
```

or

In [16]:
```
num1 = int(input("Enter 1st Number Here:"))
num2 = int(input("Enter 2nd Number Here:"))
num3 = int(input("Enter 3rd Number Here:"))

if ( num1 > num2 ) and (num1 > num3):
    print(num1,"is the largest number")
elif ( num2 > num1 ) and (num2 > num3):
    print(num2,"is the largest number")
else:
    print(num3,"is the largest number")
```

```
100 is the largest number
```

16.Python Program to Check Prime Number

In [19]:
```
num = int(input("Enter a Number here:"))

if num == 1:
    print("It is not a prime number")

if num > 1:
    for i in range(2,num):
        if num % i == 0:
            print("It is not a prime number")
        else:
            print("It is a prime number")
```

```
It is not a prime number
It is not a prime number
It is a prime number
It is a prime number
```

or

In [24]:
```python
num = int(input("Enter a Number here:"))

if num == 1:
    print("It is not a prime number")

if num > 1:
    for i in range(2,num):
        if num % i == 0:
            print("It is not a prime number")
            break
    else:
        print("It is a prime number")
```

```
It is a prime number
```

17.Python Program to Generate a Random Number

In [32]:
```python
import random

num = random.randint(0,10)

print(num)
```

```
7
```

18.Python Program to Print all Prime Numbers in an Interval

In [37]:
```python
lower = int(input("enter lower limit here:"))
upper = int(input("enter lower limit here:"))

for num in range (lower, upper + 1):
    if num > 1:
        for i in range (2,num):
            if num%i == 0:
                break
        else:
            print(num)
```

```
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47
```

19. Python Program to Convert Celsius To Fahrenheit

In [40]:
```
celsius = int(input("enter temperature in celsius:"))

fahrenheit = (celsius *(9/5))+32

print("The converted value is",fahrenheit,"fahrenheit")
```

```
The converted value is 104.0 fahrenheit
```

20.Python Program to Find the Factorial of a Number

Soultion-1: Using For Loop

In [46]:
```
num = int(input("Enter a number here:"))

fact = 1
if num < 0:
    print("Factorial of 0 does not exist")
if num == 0:
    print("Factorial of 0 is", 1)
if num > 0:
    for i in range (1,num+1):
        fact = fact * i
print("The Factorial of the given number is ", fact)
```

```
The Factorial of the given number is  24
```

Solution-2: Using Recursion

In [49]:
```
def fact(a):
    if a == 0:
        return 1
    else:
        return ((a)*fact(a-1))

num = int(input("Enter a number here: "))

result = fact(num)
print("The factorial of the given number is", result)
```

```
The factorial of the given number is 24
```

In [ ]: